



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Lipid metabolism Part I: metabolism of cholesterol and bile acids

**Zhang, Dong PhD,
Professor of Cancer Biology
Biomedical Sciences Department
dzhang12@NYIT.EDU**

**Harvey and Ferrier: Lippincott Biochemistry 8th Edition: Chapter 18
Marks' Basic Medical Biochemistry, 6e, Chapter 32, 34**

Office of Academic Affairs

Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Learning Objectives

1. Describe the important structural features of cholesterol and cholesterol esters and their physiological functions.
2. Describe the key steps of cholesterol biosynthesis and the enzymes that catalyze them.
3. Describe the mechanisms by which cholesterol biosynthesis is regulated by energy availability, hormones, food intake, and pharmacological manipulation.
4. Explain the effect of up-regulating or down-regulating plasma cholesterol levels on the intracellular biosynthesis of cholesterol and the transcriptional regulation of genes that are involved in cholesterol homeostasis.
5. Describe the biosynthesis, degradation, and regulation of bile acids; explain the clinical relevance related to bile acids metabolism.

Structure and function of cholesterol

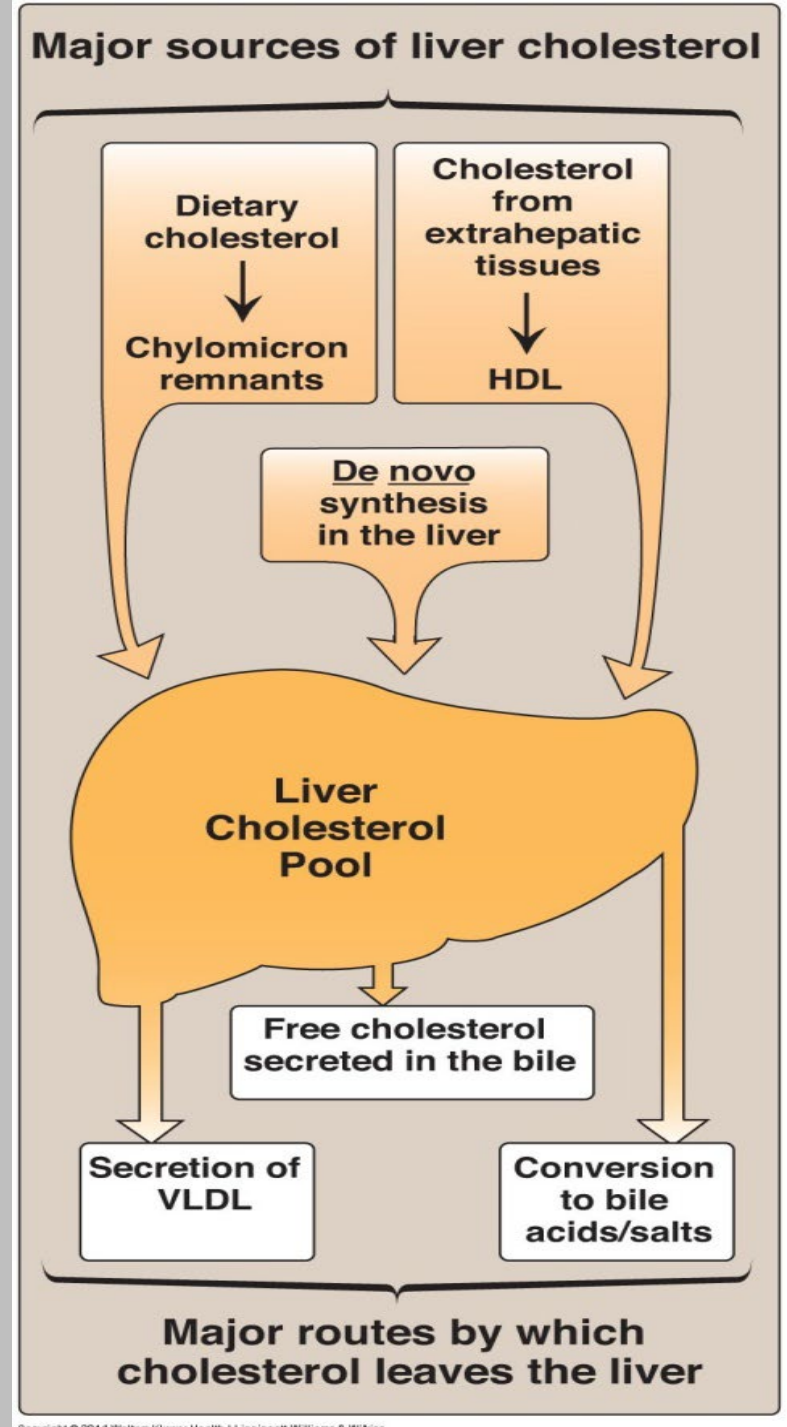
Overview of cholesterol

Main physiological functions of cholesterol:

1. A structural component of cellular membranes – modulating membrane **fluidity***
2. Precursors of:
 - Vitamin D
 - Sterol hormones
 - Bile acids

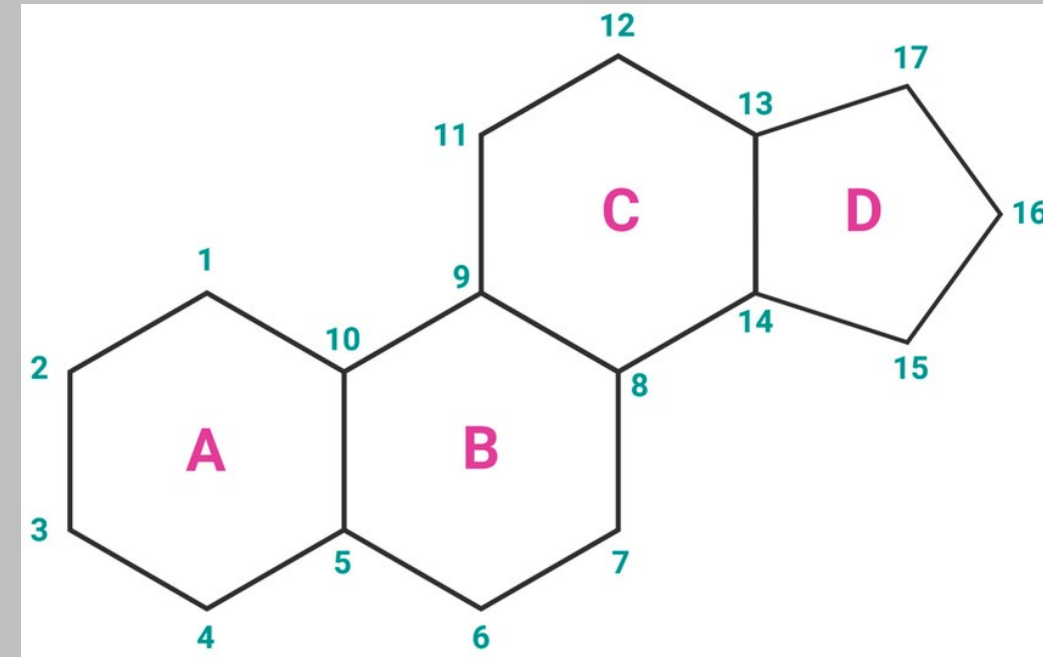
Liver plays a central role in cholesterol homeostasis

*: **Lipid bilayer** is the key feature of all cellular membranes. These thin sheets are composed largely of phospholipids. They are also in a dynamic and **fluid-like** state, meaning that their components and “fluidity” can change.



Steroid Nucleus

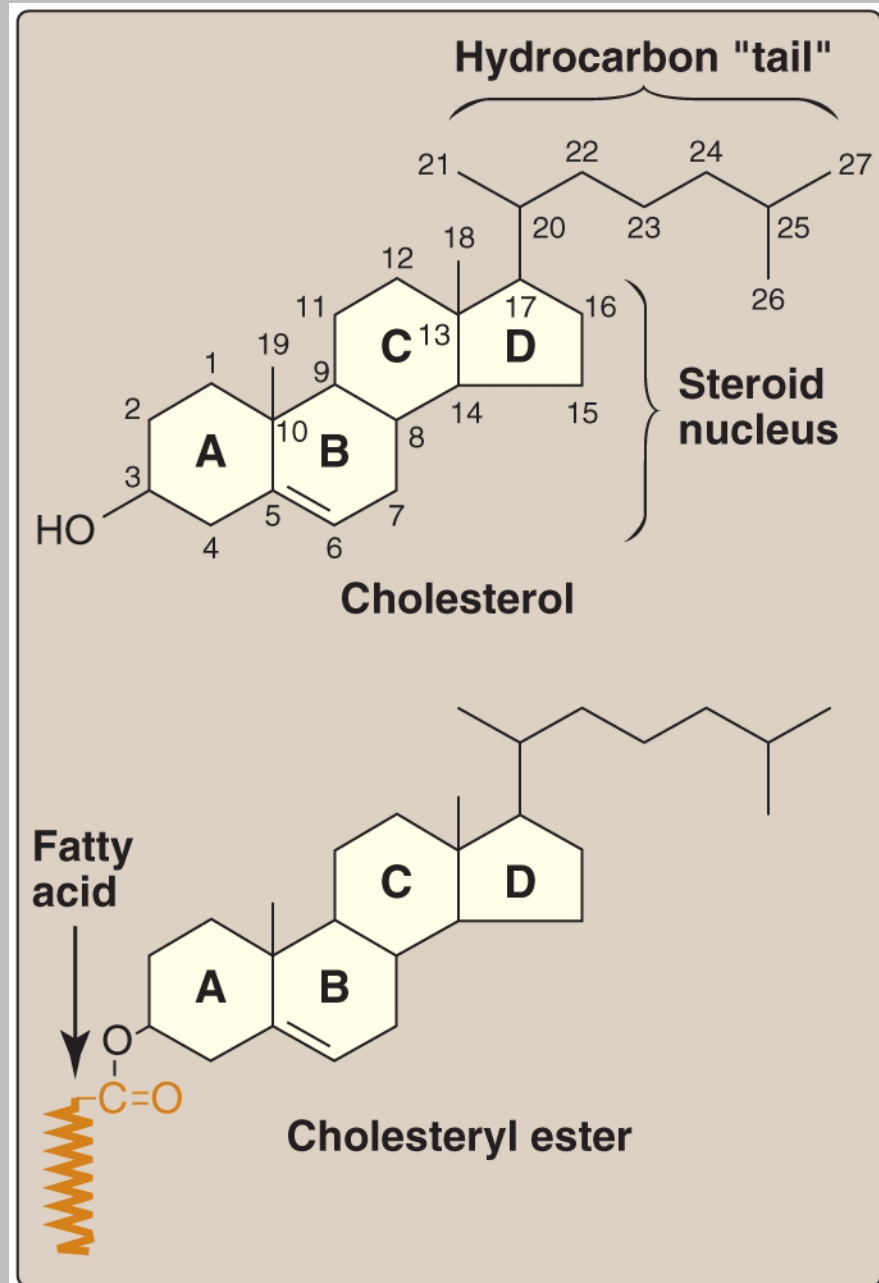
- Steroids and cholesterol are derivatives of a fused, **four ring system (A to D)**, called **Steroid Nucleus**.
- The carbons are numbered as shown.



**You DO NOT need to
memorize the structure!!!**

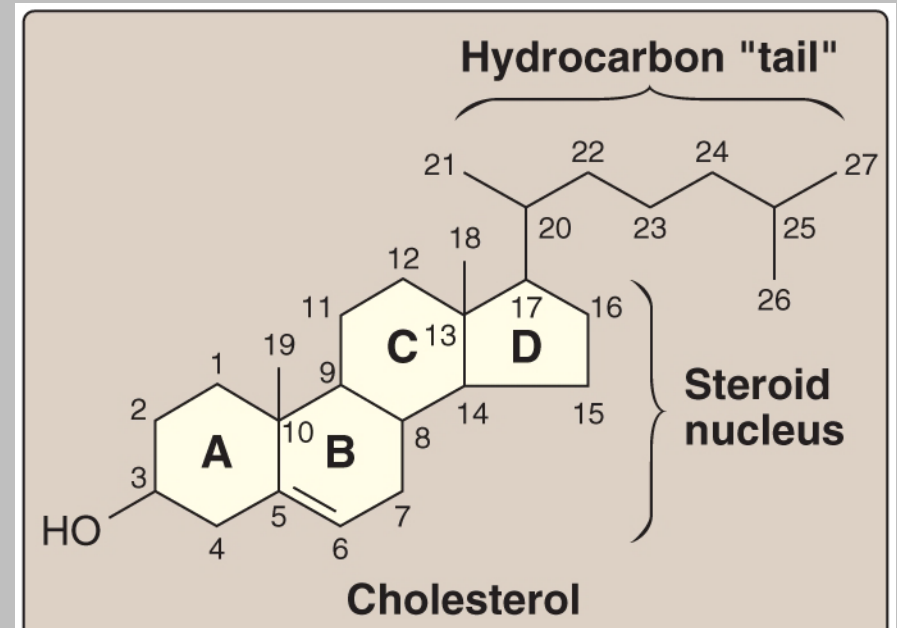
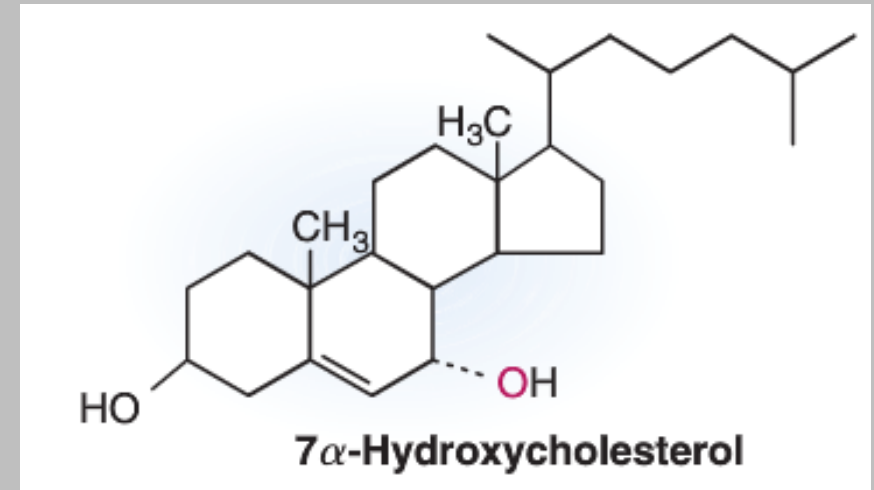
Sterols

- **Sterols** are a subclass of steroids that always contain a **hydroxyl group** at carbon 3 and an **aliphatic chain** at least eight carbons long attached to carbon 17
- The hydroxyl group at carbon 3 can be **esterified** to produce sterol esters. Cholesterol has a **double bond** at carbon 5



Nomenclature of Steroids and Sterols:

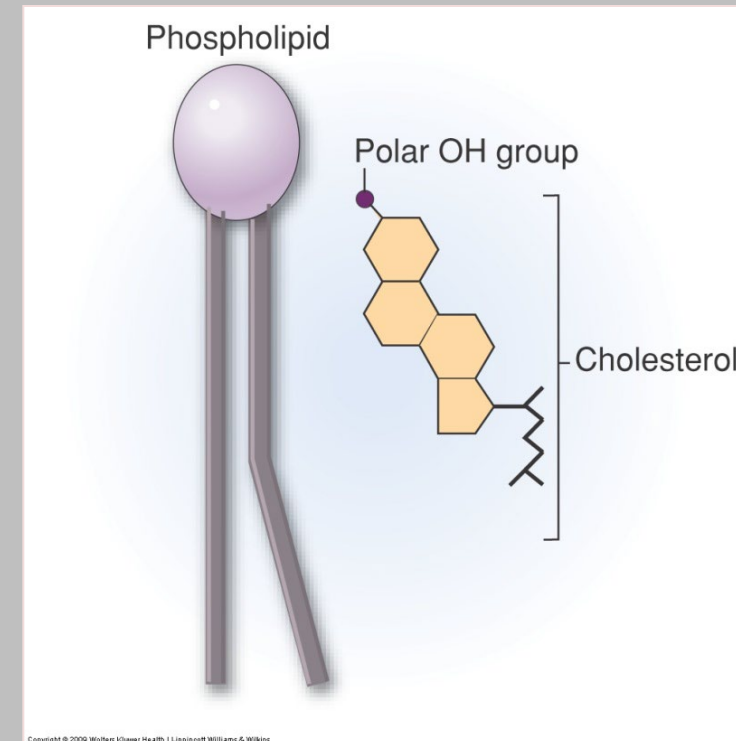
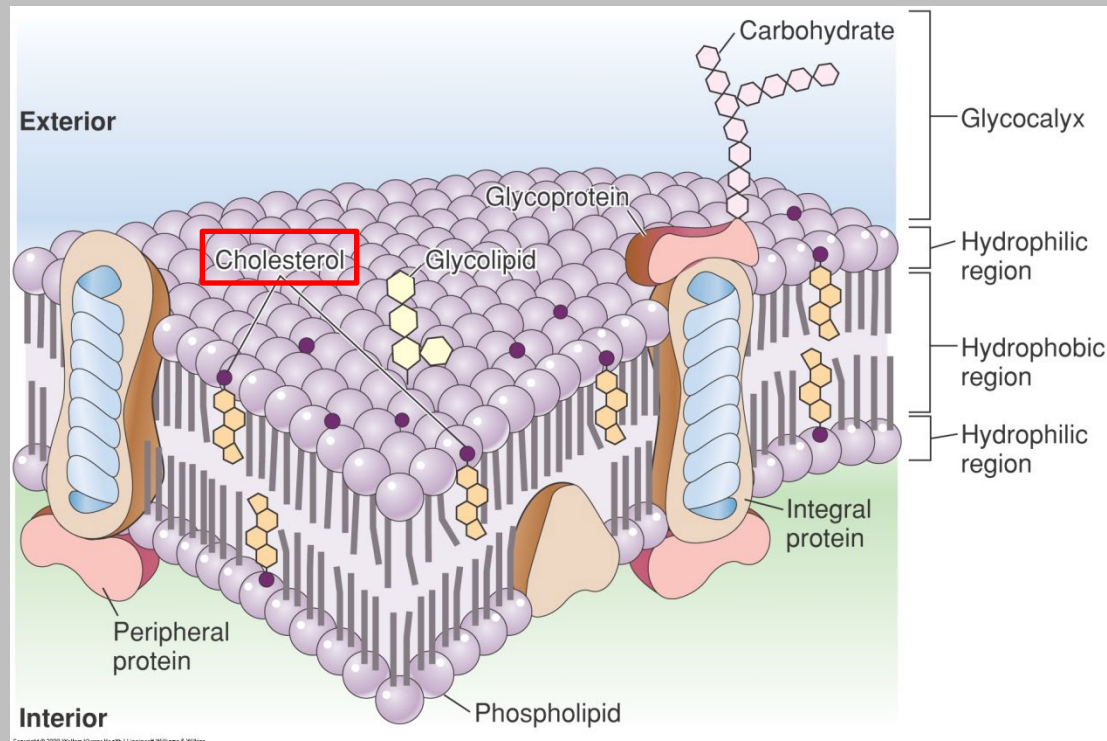
- Substituents oriented below the plane of the ring system are said to be in the α orientation. Groups in α -orientation are shown connected to the ring system by **dashed lines**.
- Substituents that are above the plane of the ring system are said to be in β orientation. β substituents are shown connected to the ring system by a **solid line**. The methyl groups in steroids, carbon 18 and 19 are always in β orientation.
- Double bonds are indicated by a Δ symbol followed by a superscript that denotes the number of the carbon atom where the double bond is located, for example, Δ^5 for cholesterol.



Functions of cholesterol-I: **membrane component**

Cholesterol, which is interspersed among membrane phospholipids, helps to maintain the **membrane fluidity**:

- is an important component of mammalian **plasma membranes** and **myelin** (about 25% of total lipid); (present at much lower concentrations in organelle membranes);
- buffers the **fluidity** of cell membranes



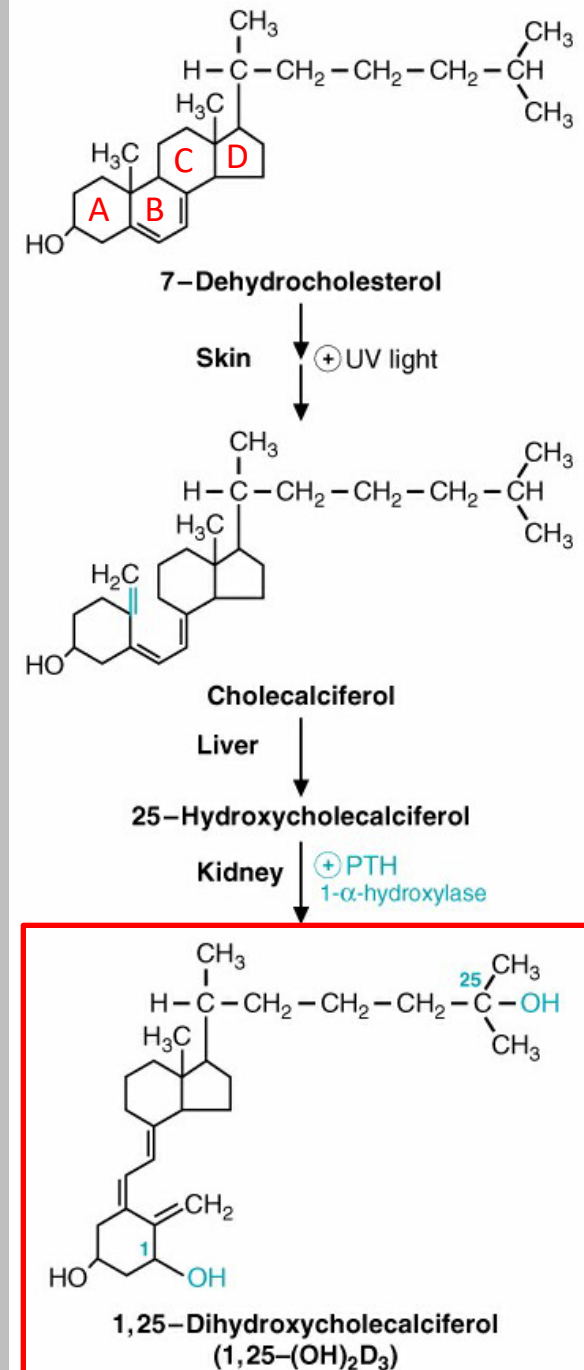
Functions of cholesterol-II:

precursor for Vitamin D biosynthesis

Vitamin D₃ and calcium metabolism:

- Vitamin D₃ is converted to a hormone that enhances intestinal Ca²⁺ absorption and demineralization of the bone
- Vitamin D₃ is synthesized endogenously from cholesterol in skin when exposed to UV
- 1,25-Dihydroxycholecalciferol is the active form of Vitamin D₃

PTH: parathyroid hormone



Functions of cholesterol-III: precursor for steroid hormone biosynthesis

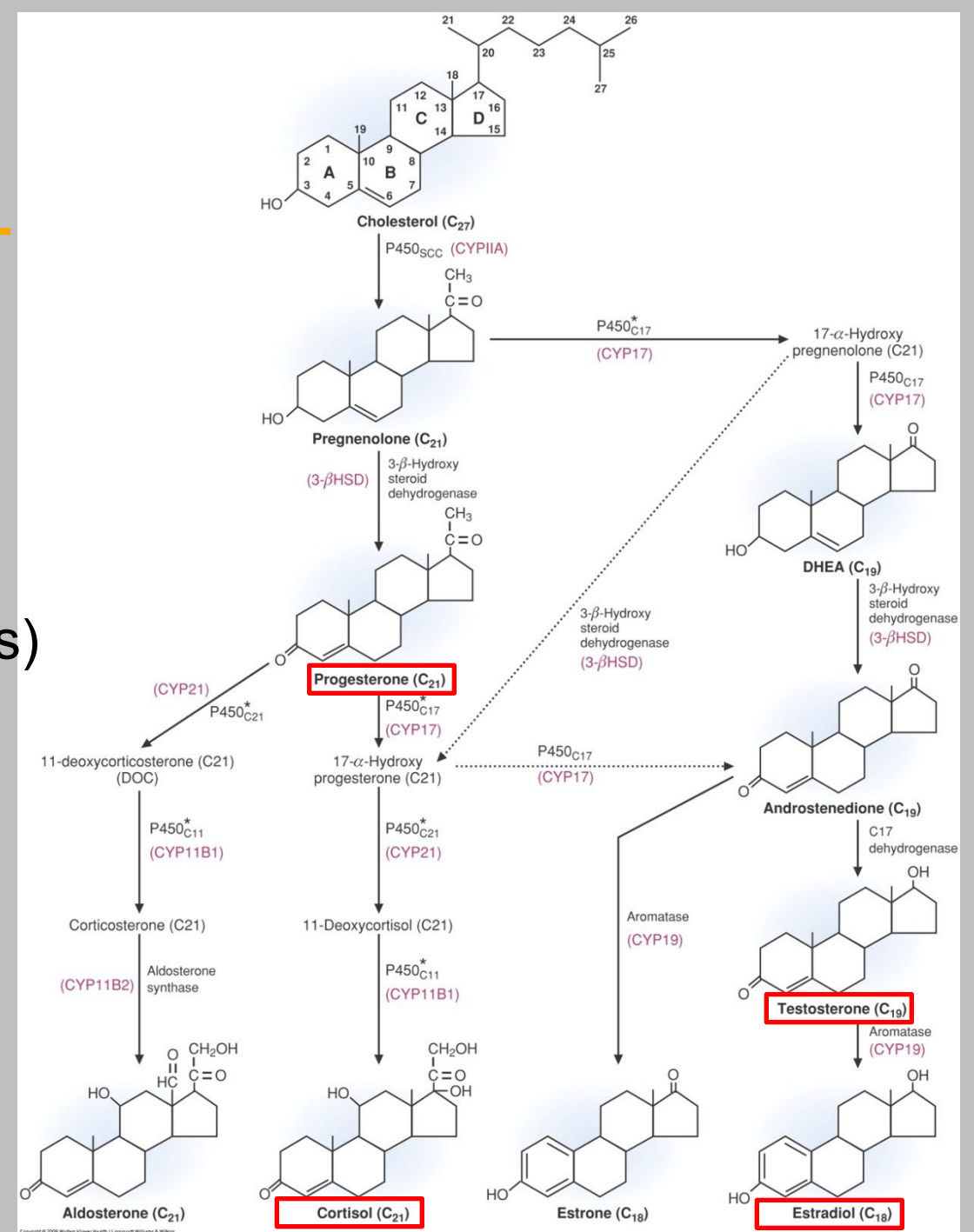
Synthesis:

- Shortening the hydrocarbon chain
- Oxidation of the steroid nucleus

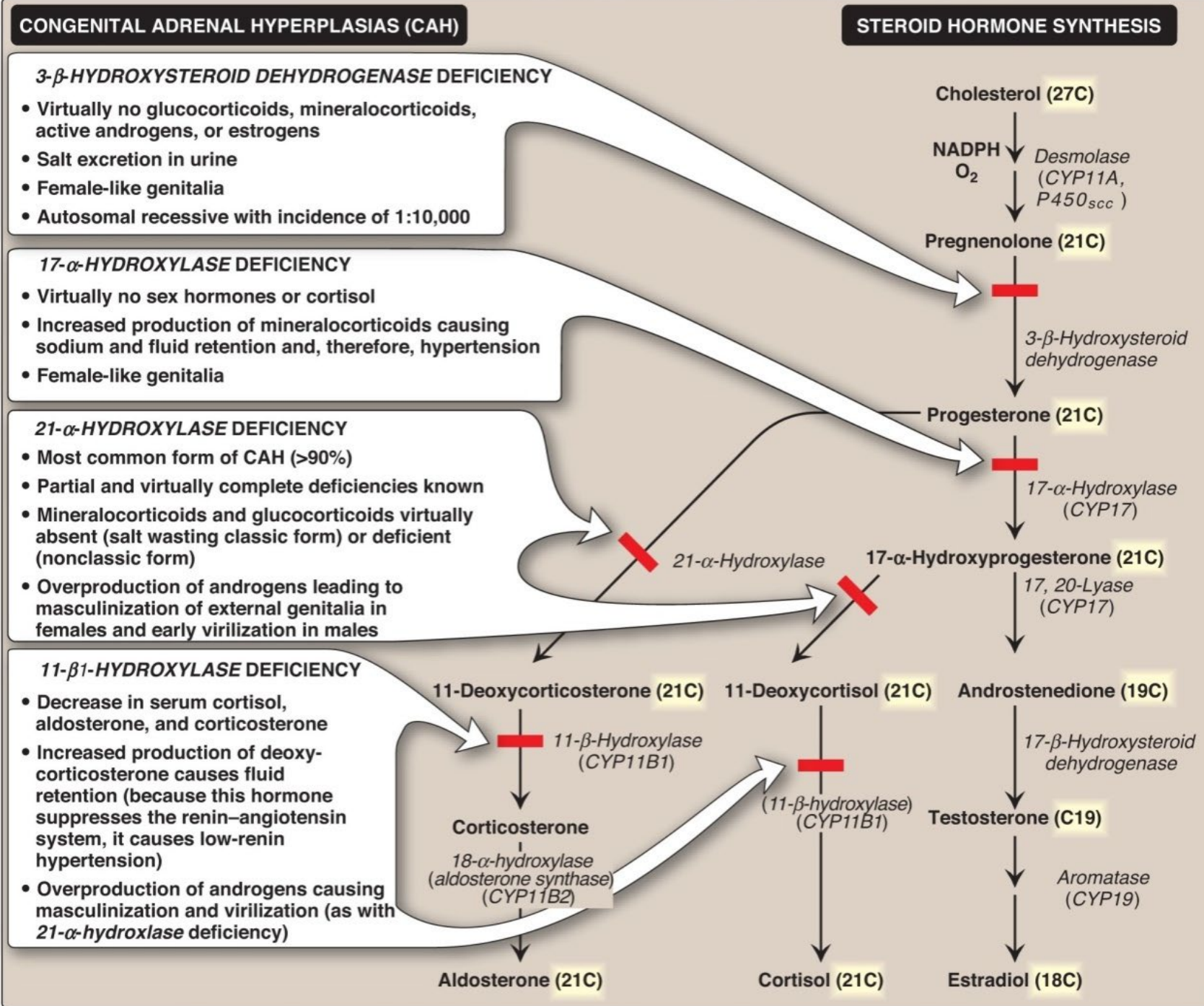
Steroid hormones:

(sex steroids, glucocorticoids, mineralocorticoids)

- Progesterone
- Testosterone
- Estradiol
- Cortisol
- Others



Genetic diseases related to steroid hormone biosynthesis

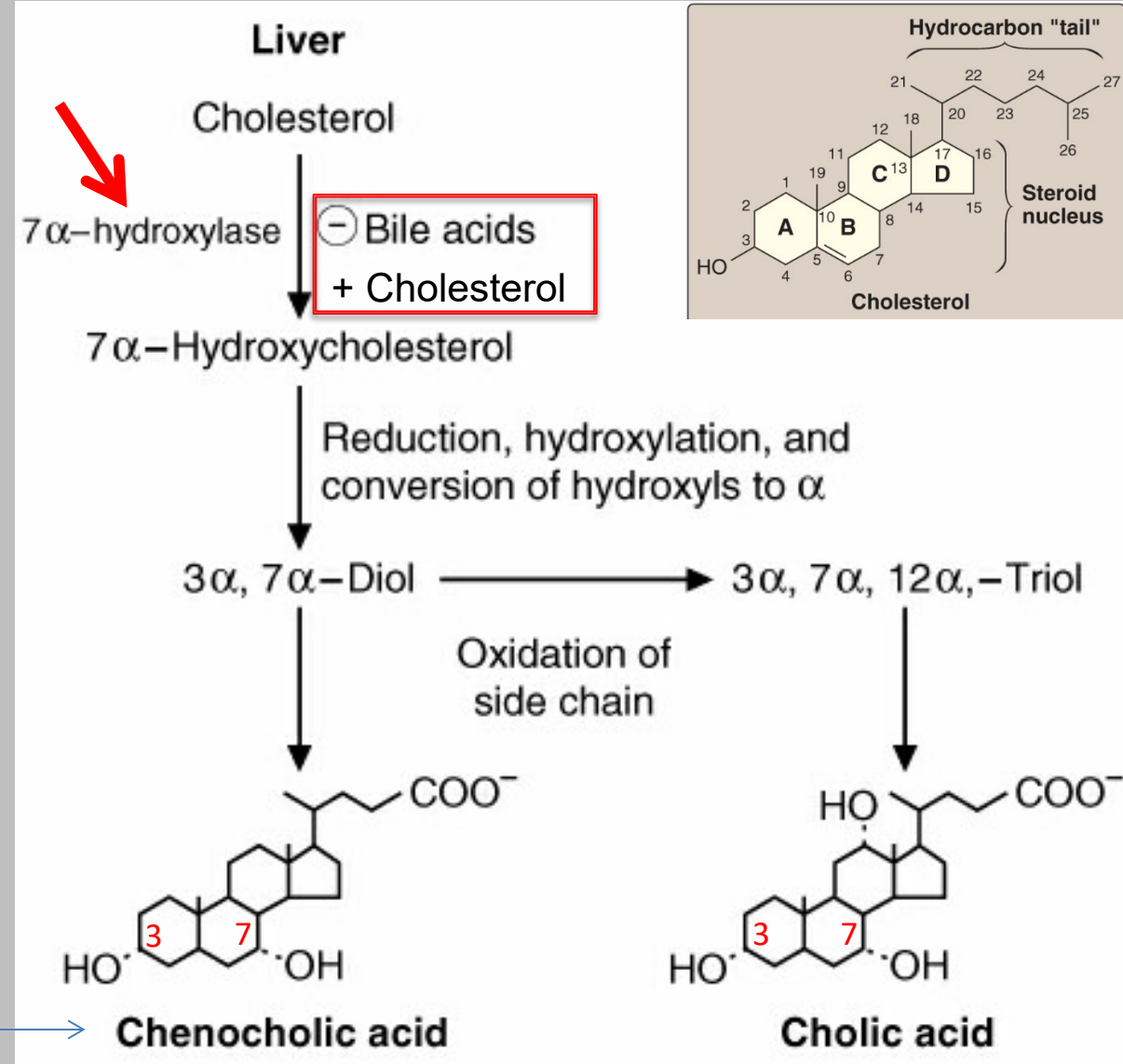


Functions of cholesterol-IV:

precursor for bile acids/salts biosynthesis

Quantitatively, the most important product of cholesterol are **bile acids**.

7-hydroxylase catalyzes the **rate-limiting step** of bile acid synthesis.



Primary bile acids:



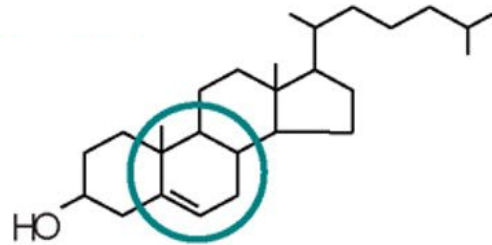
Plant Sterols

- Plants do not have significant amounts of cholesterol. But they do have other types of sterols. including **stigmasterol**, **sitosterol**, **sitosteranol**.
- Plant sterols are not absorbed efficiently by the human digestive system: after entering the enterocytes, they are actively transported back into the intestinal lumen, bring some **cholesterol** with them. Therefore, **plant sterols in the diet can decrease the absorption of cholesterol in the diet.**

Structural comparison between cholesterol and plant sterols

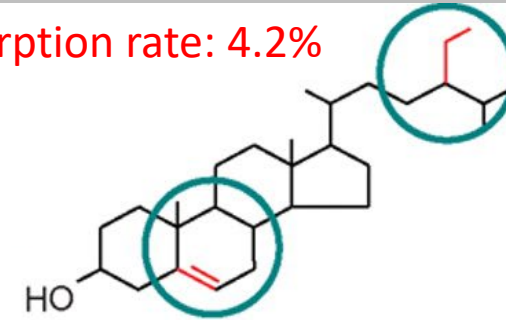
Cholesterol

Absorption rate: 33%



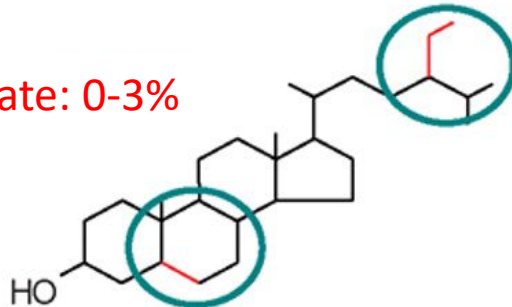
sitosterol

Absorption rate: 4.2%

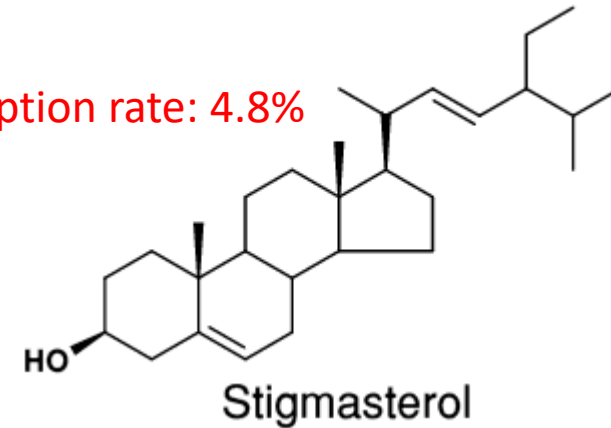


sitostanol

Absorption rate: 0-3%

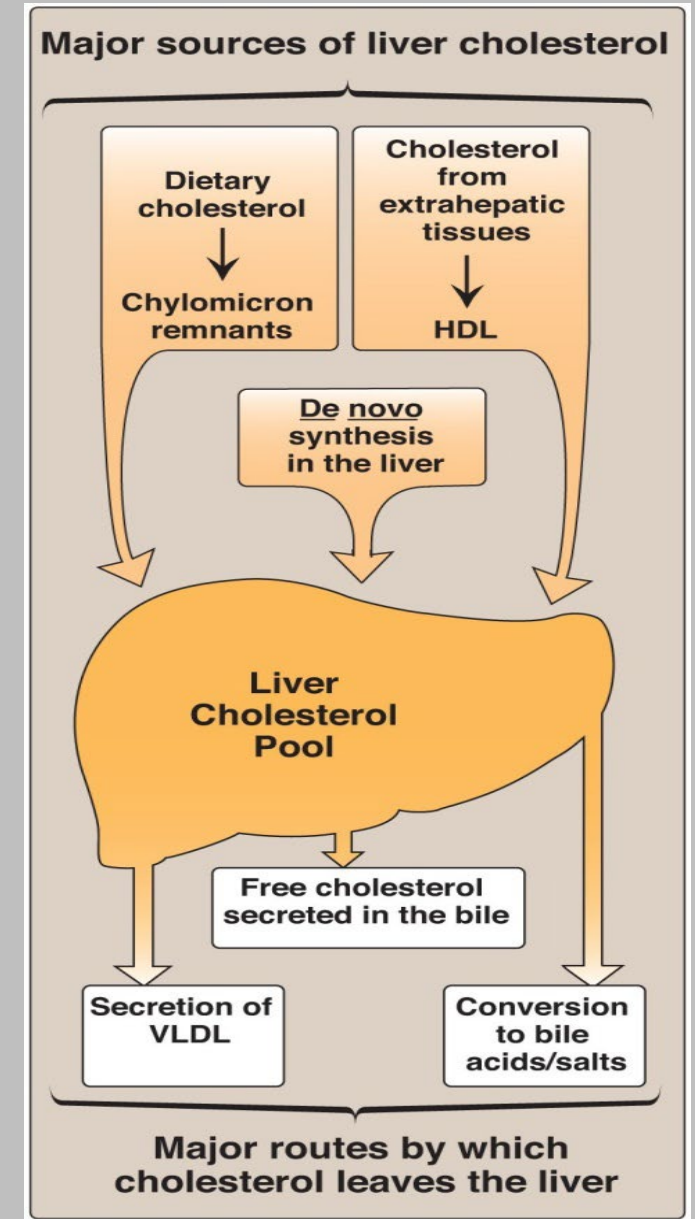


Absorption rate: 4.8%



Cholesterol Balance

- A typical Western diet contains roughly **400-600 mg /day** of cholesterol; about half is absorbed, which is mediated by the Niemann-Pick C1-like 1 protein (**NPC1L1**), the target of **ezetimibe**.
- About **800 mg / day** of cholesterol is synthesized by the body.
- This production is balanced mostly by the excretion of **bile acids**, about **1000-1100 mg/ day**.
- So, if we want to decrease cholesterol levels in the body, we could:
 1. Decrease **dietary** cholesterol uptake
 2. Decrease **cholesterol biosynthesis**
 3. Increase the **excretion** of bile acids



Biosynthesis of cholesterol



The structure of my lectures related to key metabolic pathways

- 1) What is the **purpose** or **importance** of the pathway?
- 2) What is (are) the **committed**, or **rate-limiting**, or **irreversible** reaction(s)? The **enzyme**(s)?
- 3) What are the **featured** reactions of the pathway and enzymes catalyzing the reactions?
- 4) How is the pathway **regulated**?
- 5) **Clinical correlation.**

Biosynthesis of cholesterol

- Most tissues can synthesize cholesterol, including **liver** (the **main source**), adrenals, gonads, intestine, placenta.
- Biosynthesis is metabolically regulated.

Biosynthesis of cholesterol:

Stage 1 - synthesis of **mevalonate** from **acetyl CoA**

- Cholesterol is synthesized in the **cytoplasm**; all the carbons are from **acetyl CoA**
- The **citrate transport system** transport excess **acetyl CoA** from mitochondrial matrix to the cytoplasm

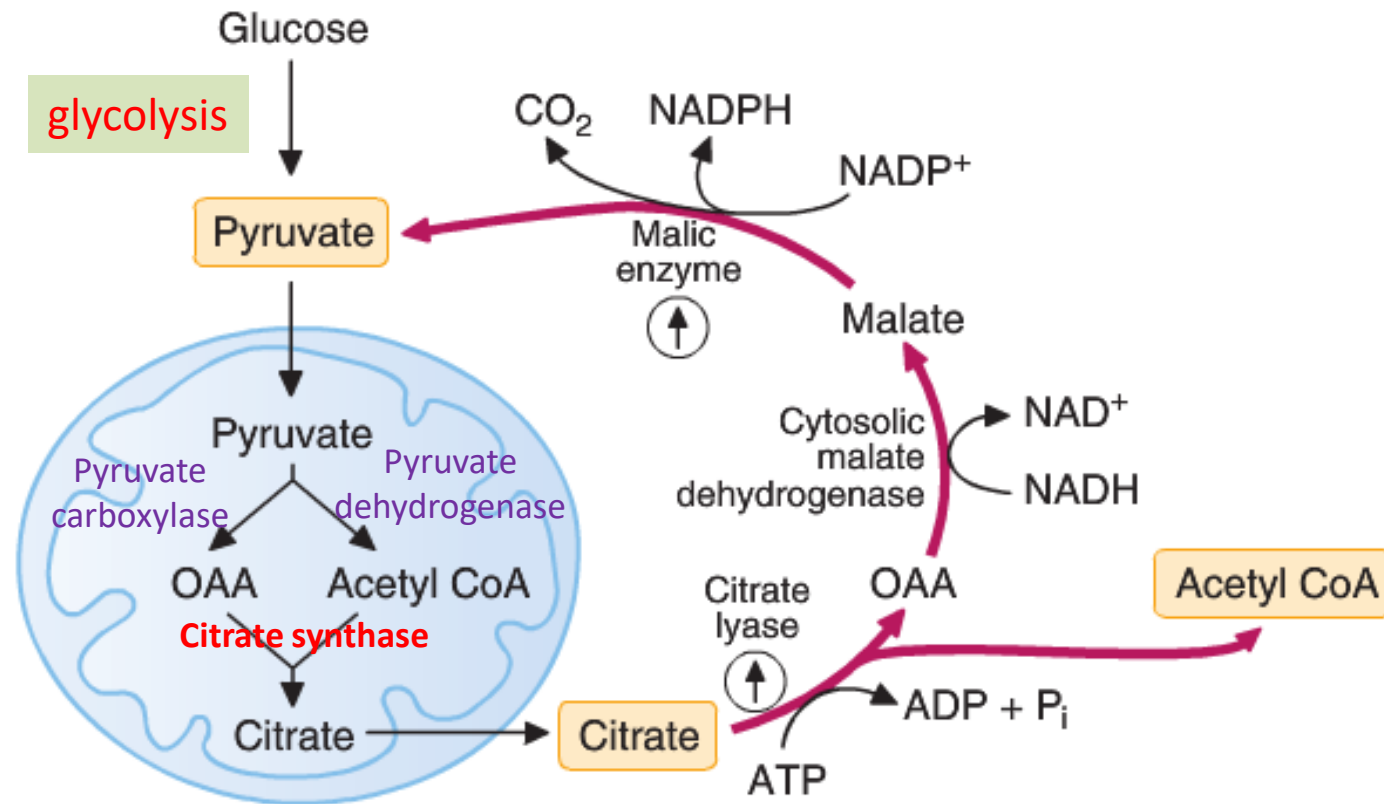


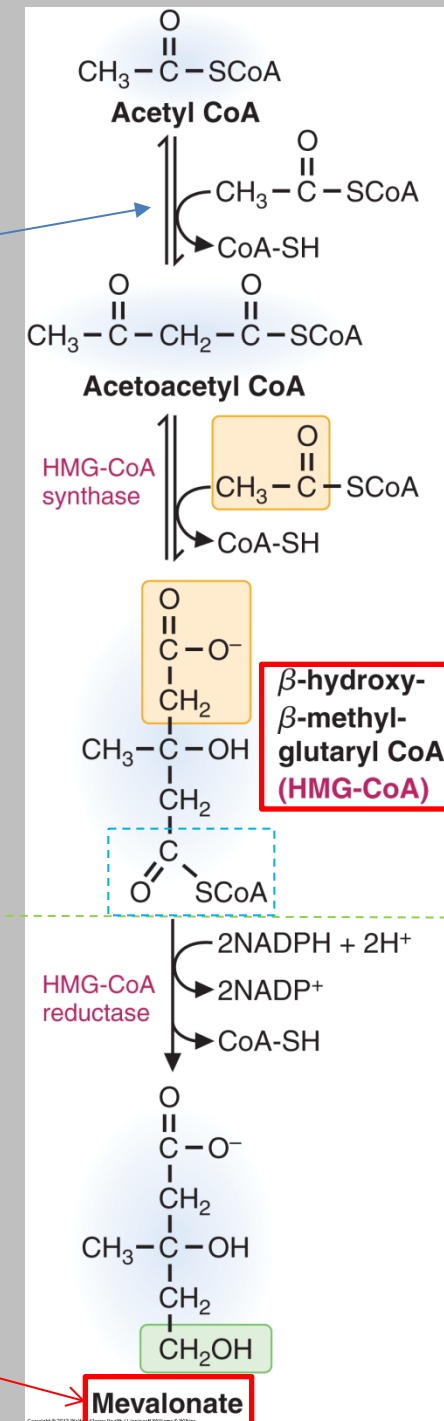
FIG. 33.5. Fate of citrate in the cytosol. Citrate lyase is also called citrate cleavage enzyme. OAA, oxaloacetate; $\uparrow$, inducible enzyme.

Biosynthesis of cholesterol: Stage 1 (continued)

Stage 1: Synthesis of **mevalonate (6C)** from **acetyl CoA (2C)**

1. Synthesis of **acetoacetyl CoA (4C)** from two molecules of **acetyl CoA** by **thiolase**
2. Cytosolic **HMG CoA synthase** combines **acetoacetyl CoA** with another molecule of **acetyl CoA**
3. **HMG CoA reductase** catalyzes the **rate-limiting step** and is the **key regulatory** point in cholesterol biosynthesis.
 - Reduces HMG CoA and produce one **mevalonate**
 - Requires **2X NADPH**
 - Located on the **ER** membrane
 - **Irreversible**

3X Acetyl CoA



IPP is the activated 5-carbon unit

You DO NOT need to memorize this!!!



The fates of cholesterol

The cholesterol may then be esterified :

- Transfer of acyl groups from fatty acyl CoA by **acyl CoA- cholesterol acyltransferase (ACAT)** and stored in the cell as cholesterol esters
- Secreted out as a part of lipoproteins (e.g., **VLDL**)
- Converted to cholesterol esters by the transfer of an acyl group from carbon 2 of lecithin (also called Phosphatidylcholine) by **Lecithin - cholesterol acyltransferase (LCAT)** in **HDL** in the plasma.

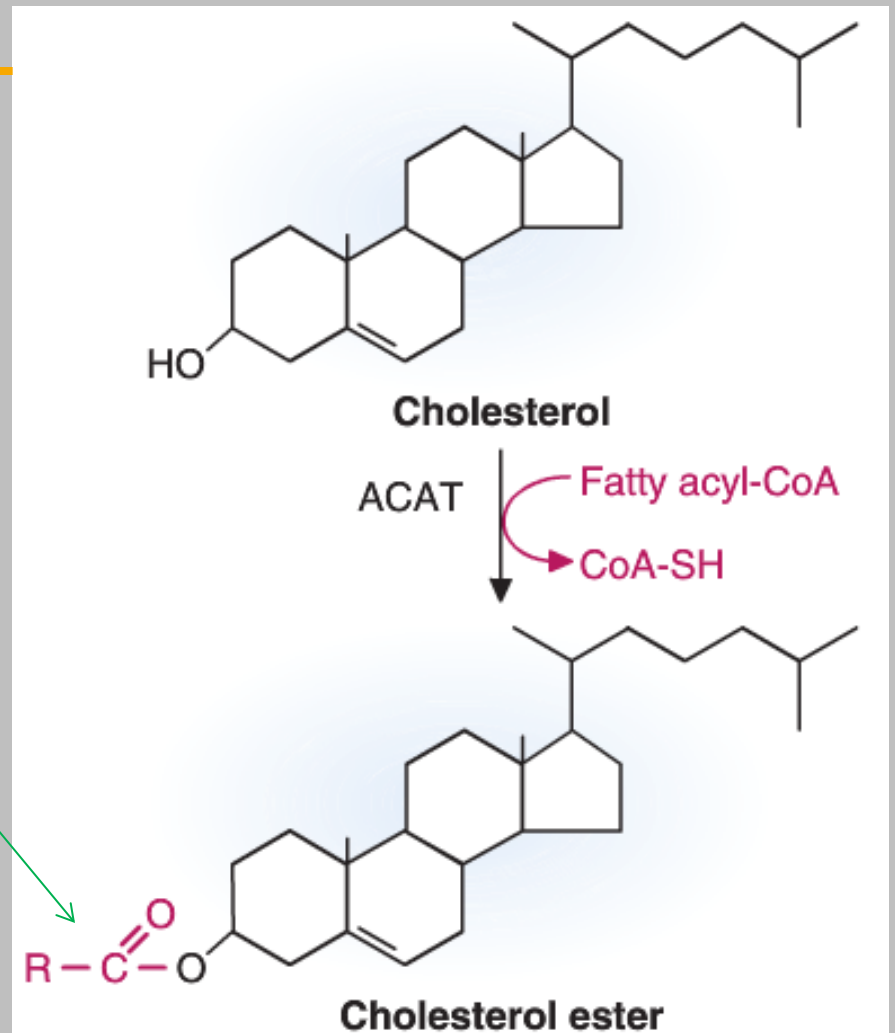
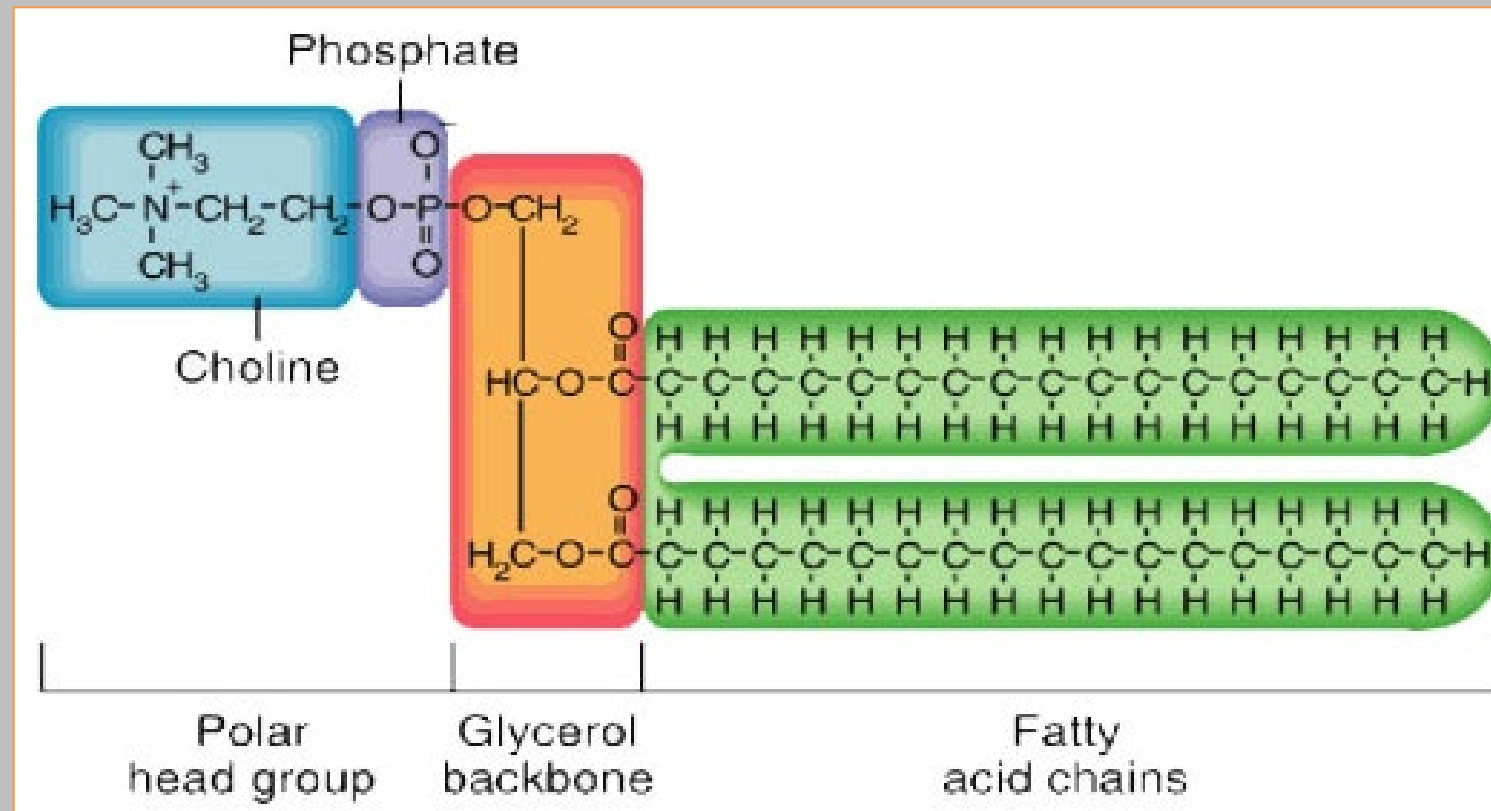


FIG. 34.8. The ACAT reaction, producing cholesterol esters. ACAT, acyl-CoA-cholesterol acyl transferase.

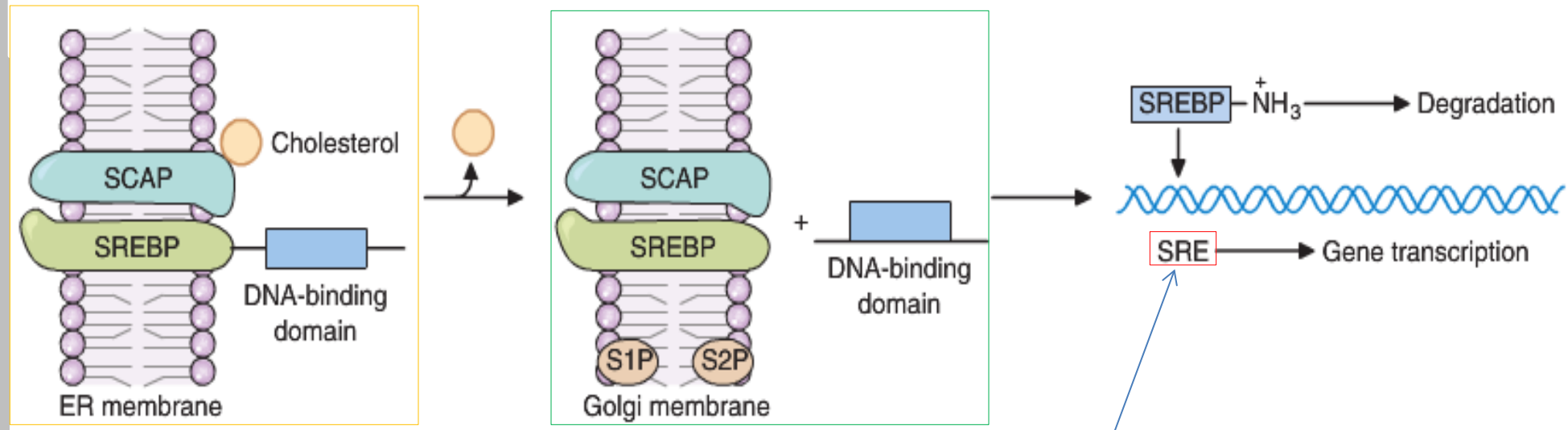
Structure of Lecithin (Phosphatidylcholine)

Lecithin is a phospholipid



Regulation of cholesterol biosynthesis

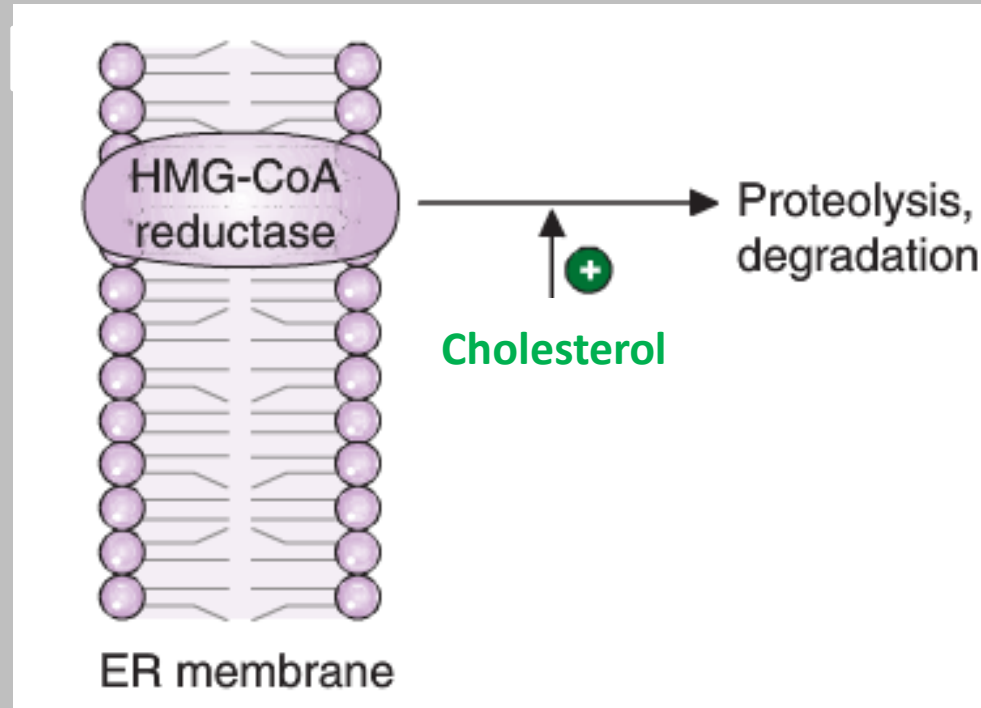
Regulation of cholesterol biosynthesis-I: via transcription



Transcriptional regulation of cholesterol biosynthesis:

- Sterol Regulatory Element Binding Protein (**SREBP**) regulates the transcription of **HMG CoA reductase**
- SREBP DNA-binding domain binds to the Steroid Response Element (**SRE**) of the HMG CoA reductase gene and turns on its transcription when cholesterol levels are low
- **SREBP** is an ER membrane protein and binds **SCAP** (SREBP Cleavage Activating Protein)
 - when [**Cholesterol**] is **high**, cholesterol binds to **SCAP**, which then traps SREBP at the ER.
 - when [**Cholesterol**] is **low**, **SCAP-SREBP** is transported to **Golgi**
- In **Golgi**, **SCAP** activates certain **proteases**, which then cleave the **SREBP** and release its DNA binding domain, which then translocates to the nucleus, binds **SRE**, and **activates** the transcription of genes involved in cholesterol biosynthesis, including HMG CoA reductase.

Regulation of cholesterol biosynthesis-II: via degradation



Regulation of cholesterol biosynthesis at the HMG CoA reductase step:

Post-transcriptional control when cholesterol is abundant:

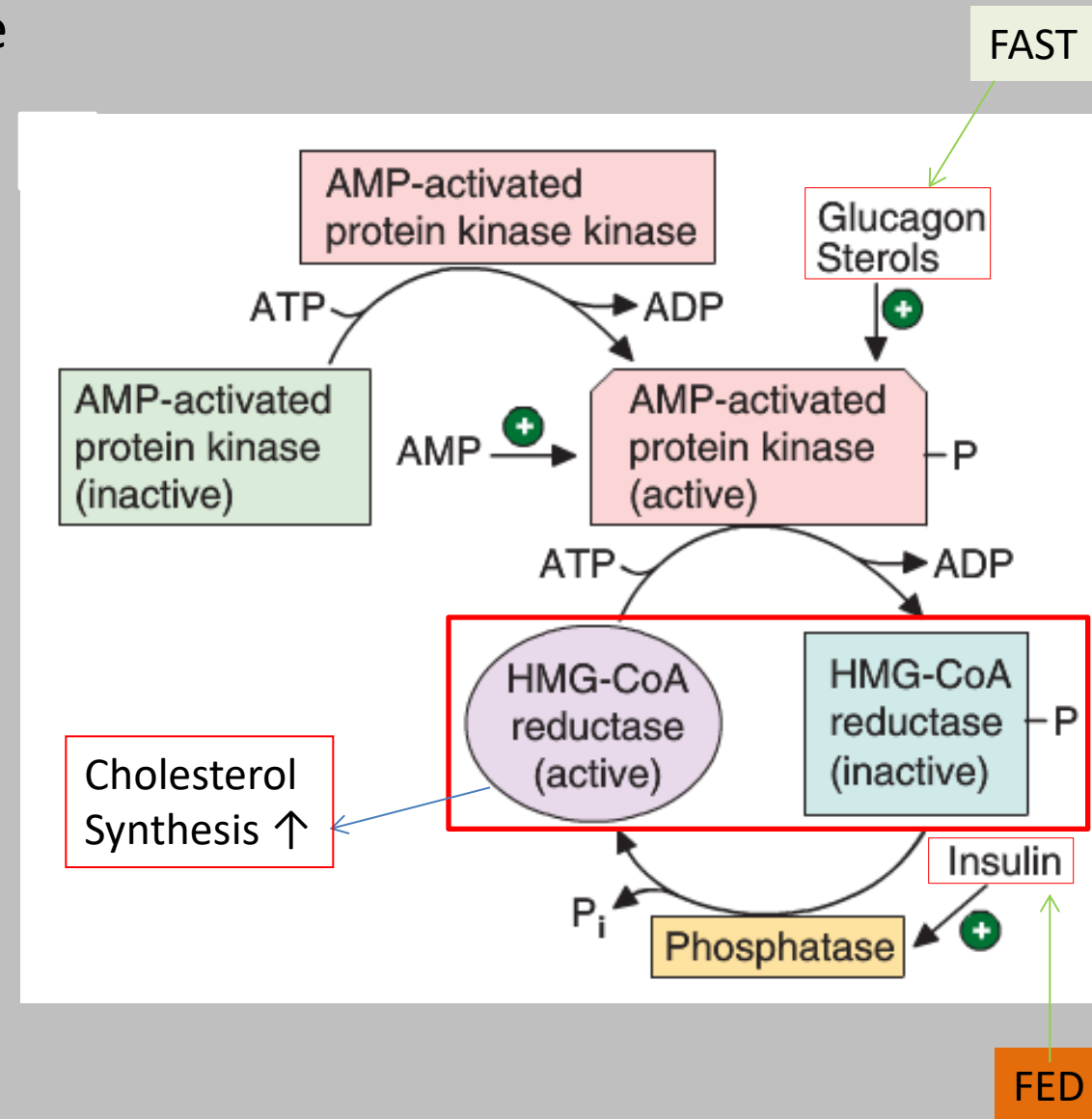
- HMG CoA reductase **mRNA** is targeted for degradation
- HMG CoA reductase **protein** is targeted for degradation by the proteasome

Regulation of cholesterol biosynthesis-III:

phosphorylation/dephosphorylation

Regulation of cholesterol biosynthesis through the phosphorylation and dephosphorylation of HMG CoA reductase:

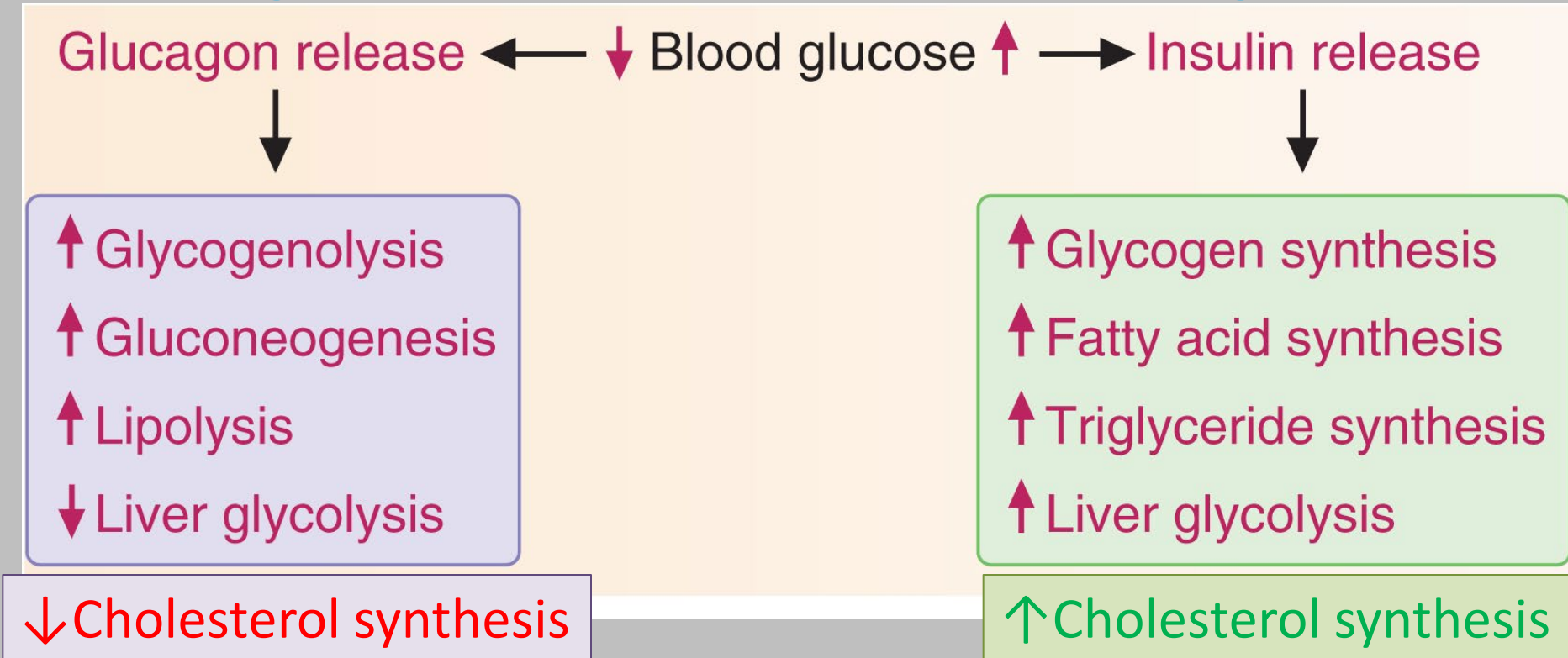
- AMP $\uparrow$ (low energy signal) $\rightarrow$ AMP-activated protein kinase (**AMPK**) $\uparrow$ $\rightarrow$ phosphorylation $\uparrow$
- **Glucagon** and high level of sterols can activate AMPK $\rightarrow$ phosphorylation $\uparrow$
- Phosphorylation of HMG CoA reductase reduces its activity.
- **Insulin** activates a protein phosphatase, resulting in the dephosphorylation of HMG CoA reductase, which increases its activity.



Hormonal regulation of metabolism

Mobilizing hormone

Storage hormone



Insulin activates a **PHOSPHATASE**

Glucagon activates a **KINASE** (PKA, AMPK)



Summary of the regulation of cholesterol biosynthesis - HMG CoA reductase

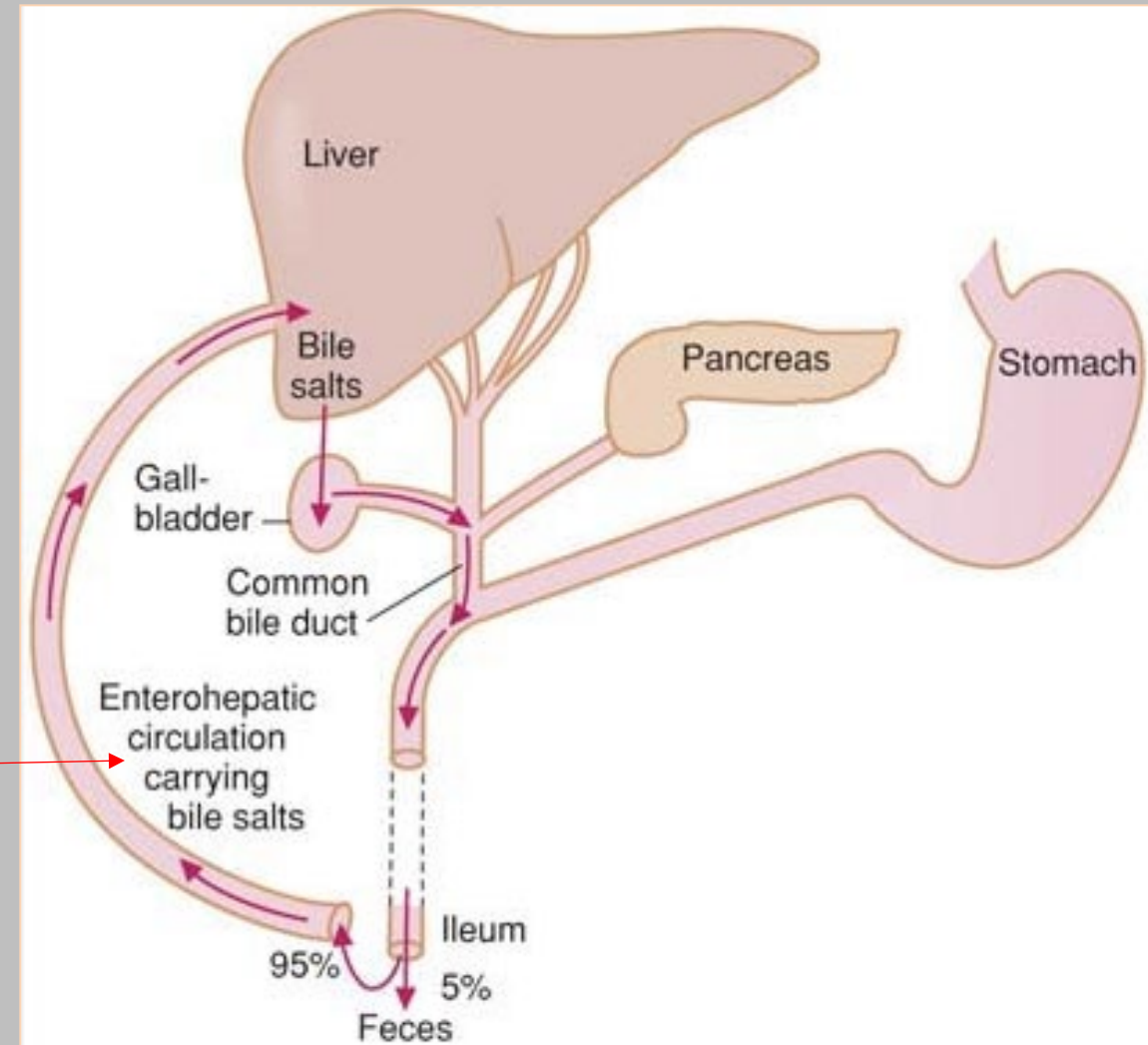
- Catalyzes the **rate-limiting step**
- **Cholesterol** functions as a **feedback inhibitor** of HMG CoA reductase synthesis through a delicate **transcriptional** regulation system: **SREBP-SCAP** is retained in the ER membranes when cholesterol level is high. When cholesterol is low, **SREBP-SCAP** is transported to Golgi; at Golgi, **SREBP** is cleaved and the cleavage fragment of **SREBP** enters the nucleus to serve as a transcription factor and stimulates the expression of HMG CoA reductase and other genes
- Both HMG CoA reductase mRNA and protein can be **degraded** by when cholesterol level is high.
- **AMP**, a low energy indicator, controls HMG CoA reductase activity through phosphorylation (by **AMPK, inactive**) and dephosphorylation (**active**).
- **Insulin** and **thyroid hormone** **stimulate** HMG CoA reductase activity; **glucagon, cholesterol, chylomicron remnants**, and **Low-density lipoproteins (LDL)** **inhibit** HMG CoA reductase activity
- HMG CoA reductase is the molecular target of cholesterol lowering drugs - **statins**

Synthesis of Bile Acids

The terms, bile acids and bile salts, are often used interchangeably!!!

The role of bile salts in the digestion and absorption of lipids

- Dietary lipids (fat) have low solubility
- Bile salts act as the **detergent** to emulsify dietary lipids
- Bile is a mixture of water, phosphatidylcholine (**lecithin**), bile salts, and cholesterol
- Cholesterol is the **precursor** of bile salts
- Bile salts are synthesized in the **liver** and stored in the gallbladder
- Bile salts can be reabsorbed in the GI tract for re-circulation back to the liver (**enterohepatic circulation**)
- A significant amount of bile salts are excreted in the feces everyday. Bile salts excretion is an important way to reduce **cholesterol** level



Synthesis of bile salts-I

Bile acids (**24C**) are **synthesized** from cholesterol (**27C**) by:

- increasing **hydroxylation** of the steroid nucleus
- removing **three carbons (-3C)** from the aliphatic chain
- adding a **carboxyl group** at carbon 24
- **reducing** the double bond in cholesterol

Hydroxylation and **carboxylation** increase the solubility of bile salts.

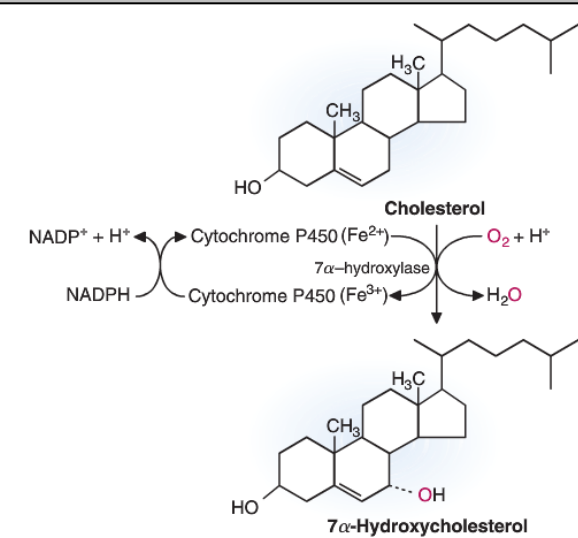
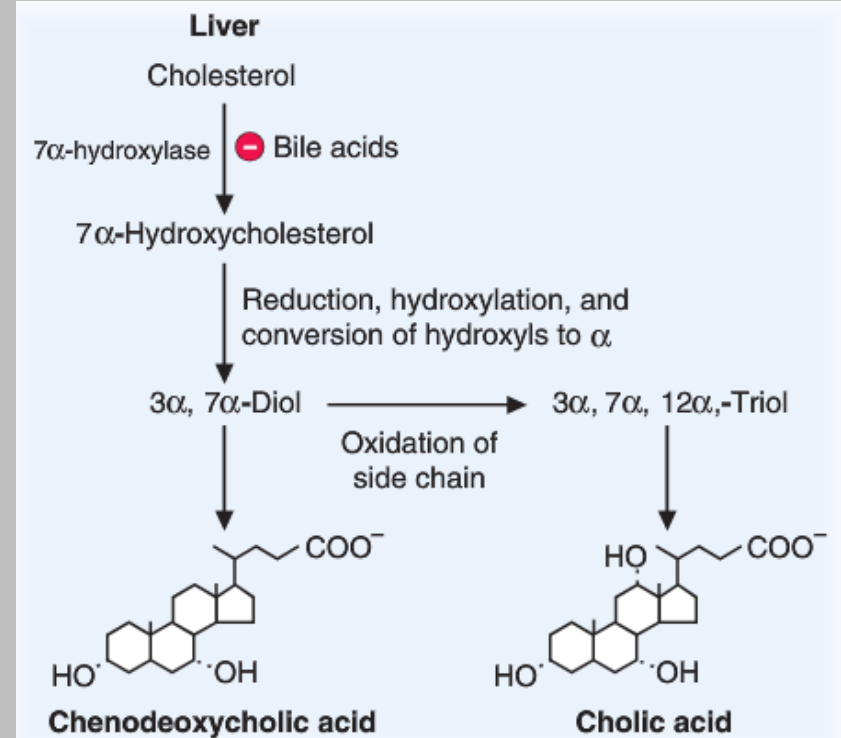


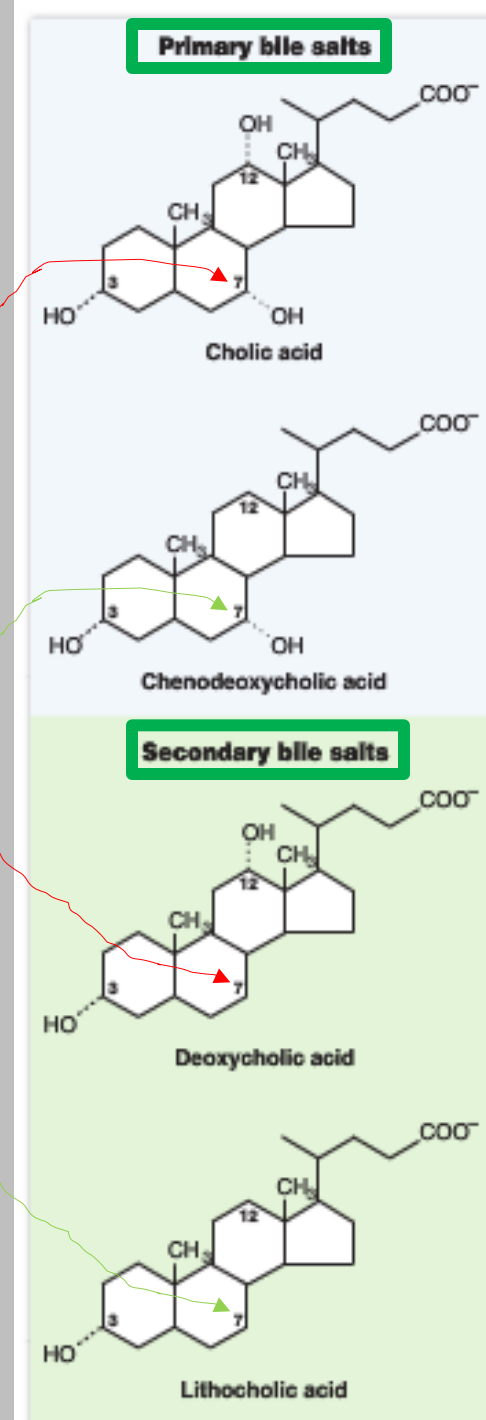
FIG. 34.9. The reaction catalyzed by 7 α -hydroxylase. An α -hydroxyl group is formed at position 7 of cholesterol. This reaction, which is inhibited by bile salts, is the rate-limiting step in bile salt synthesis.



Primary bile acids:

Synthesis of bile salts-II

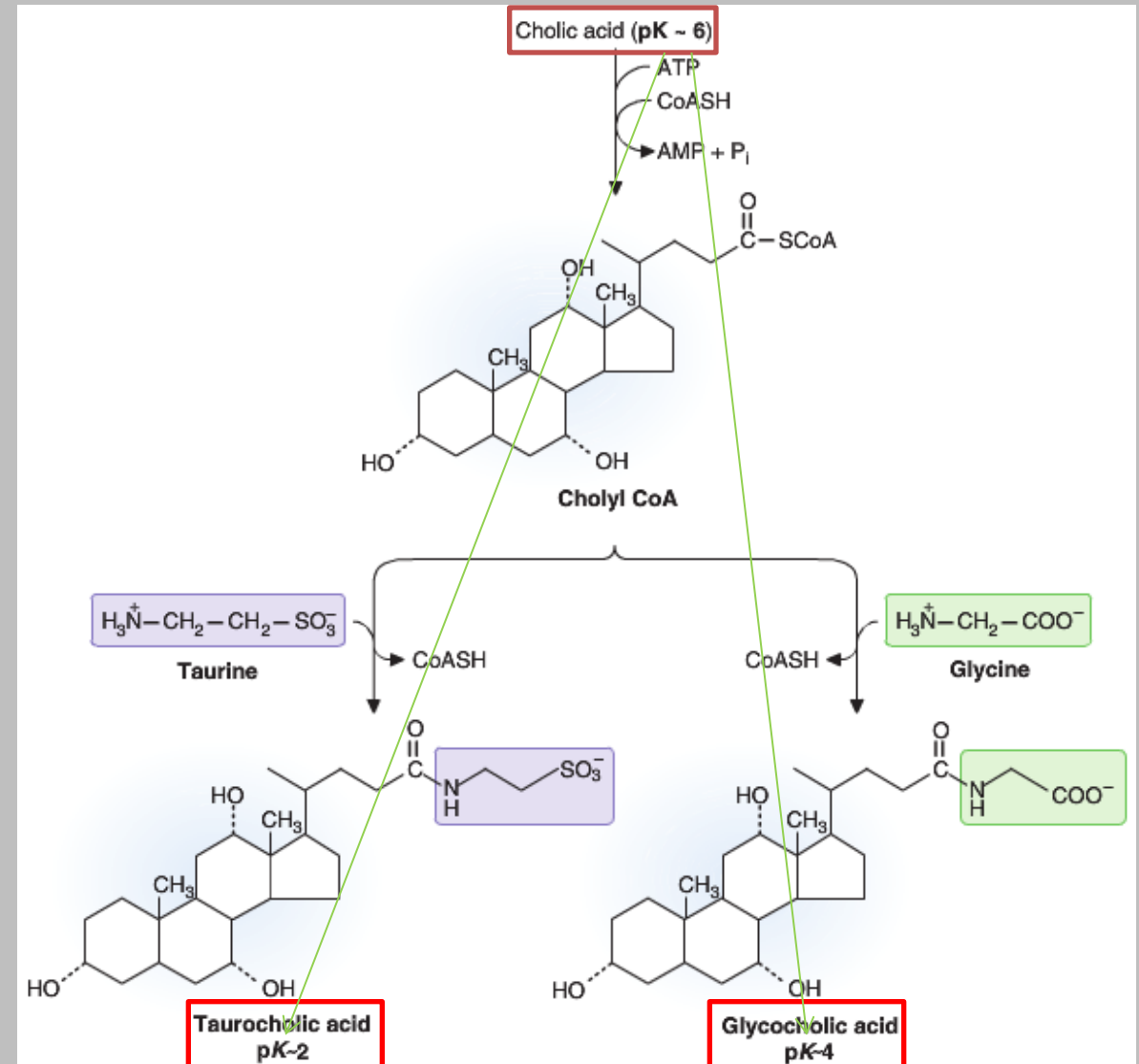
- Bile acids are synthesized in the **liver** directly from cholesterol; **cholic acid** and **chenodeoxycholic acid** are also known as the 'primary bile acids'
- Hydroxylation** at carbon 7 by **7 α -hydroxylase** is the **rate-limiting step**
- Deoxycholic acid** and **lithocholic acid** are formed in the intestine from the primary bile acids by intestinal **bacterial enzymes** and are thus called the 'secondary bile acids'
- Secondary bile salts** are **less soluble** (because they lost the OH groups at carbon-7) and more likely to be excreted



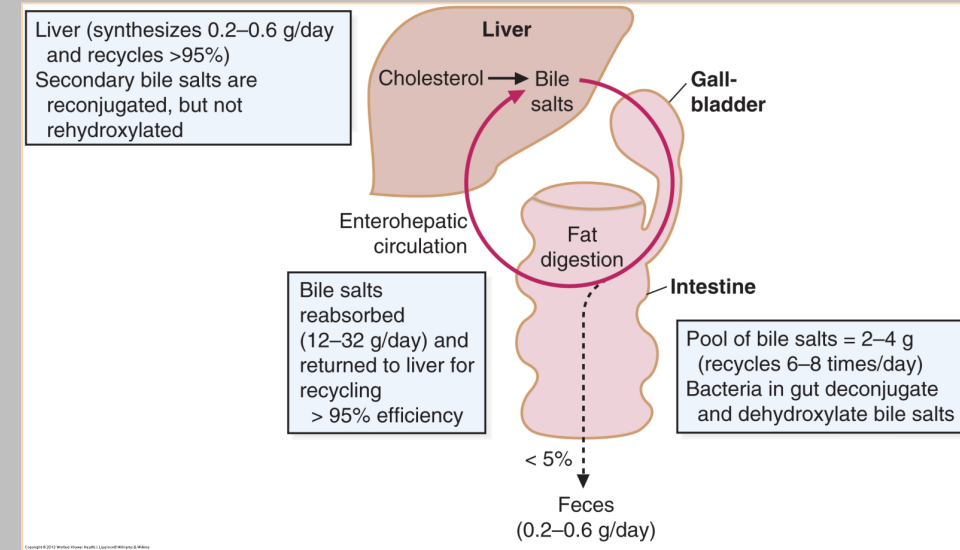
Synthesis of bile salts-III

Conjugated bile salts: the bile acids can also be conjugated to **glycine** or **taurine** via peptide bonds through the carbon 24 carboxylate.

- Conjugated bile salts have even **lower pKa**
- A higher percentage of the **conjugated bile salts** is present in the ionized form in intestine (pH ~6)
- They are **better detergents** than the unconjugated bile salts



Summary of bile salts metabolism



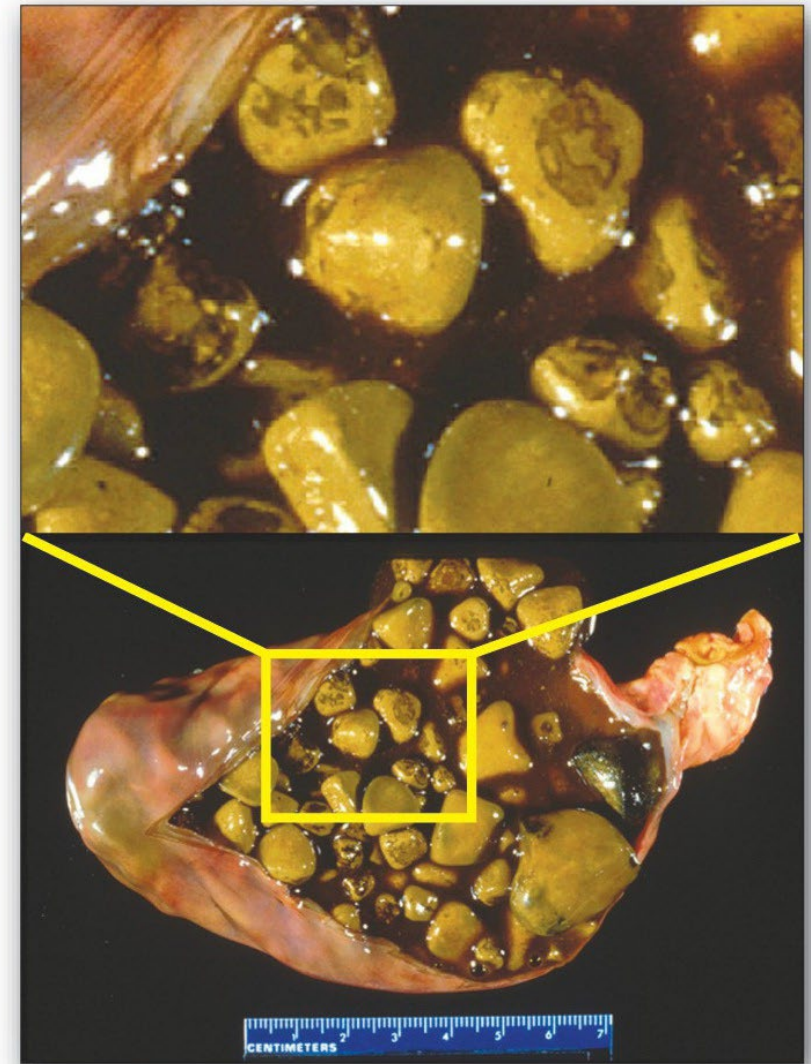
- The bile acids secreted into the intestine are efficiently re-absorbed back into the blood stream at the ileum (**> 95%**) and subsequently return to the liver.
The liver → bile duct → duodenum → ileum → portal vein → liver cycle of bile acids is called the **enterohepatic circulation**.
- The efficiency of the liver in extracting bile acids may be compromised under conditions of liver disease, resulting in increased bile acids in the blood.
- decreasing the re-absorption of bile salts can be a viable strategy to reduce the level of cholesterol, for example, by bile salts sequestrant (**cholestyramine**) and dietary fibers.

A note: In medical terms, "entero" is a prefix that means "intestine". It comes from the Greek word *énteron*, which also means "intestine".

Bile salt deficiency: cholelithiasis

Cholelithiasis (cholesterol gallstone disease):

- ❖ the imbalance between **cholesterol** (very hydrophobic) and **phospholipids** and **bile salts** (amphipathic) leads to the formation of precipitants/stones in the gallbladder.



Summary

- **Key concepts:** steriod, sterol, steroid nucleus, plant sterols, bile acids/bile salts, primary bile salts, secondary bile salts, conjugated bile salts, enterohepatic circulation
- Physiological function of cholesterol and bile acids
- The rate-limiting step of cholesterol and bile acids biosynthesis and the key enzymes involved
- Key regulatory mechanisms of cholesterol and bile acids biosynthesis
- **Clinical correlations** (disease symptoms, biochemical basis, and treatment): various strategies to lower cholesterol level; cholelithiasis

Where does acetyl CoA come from?

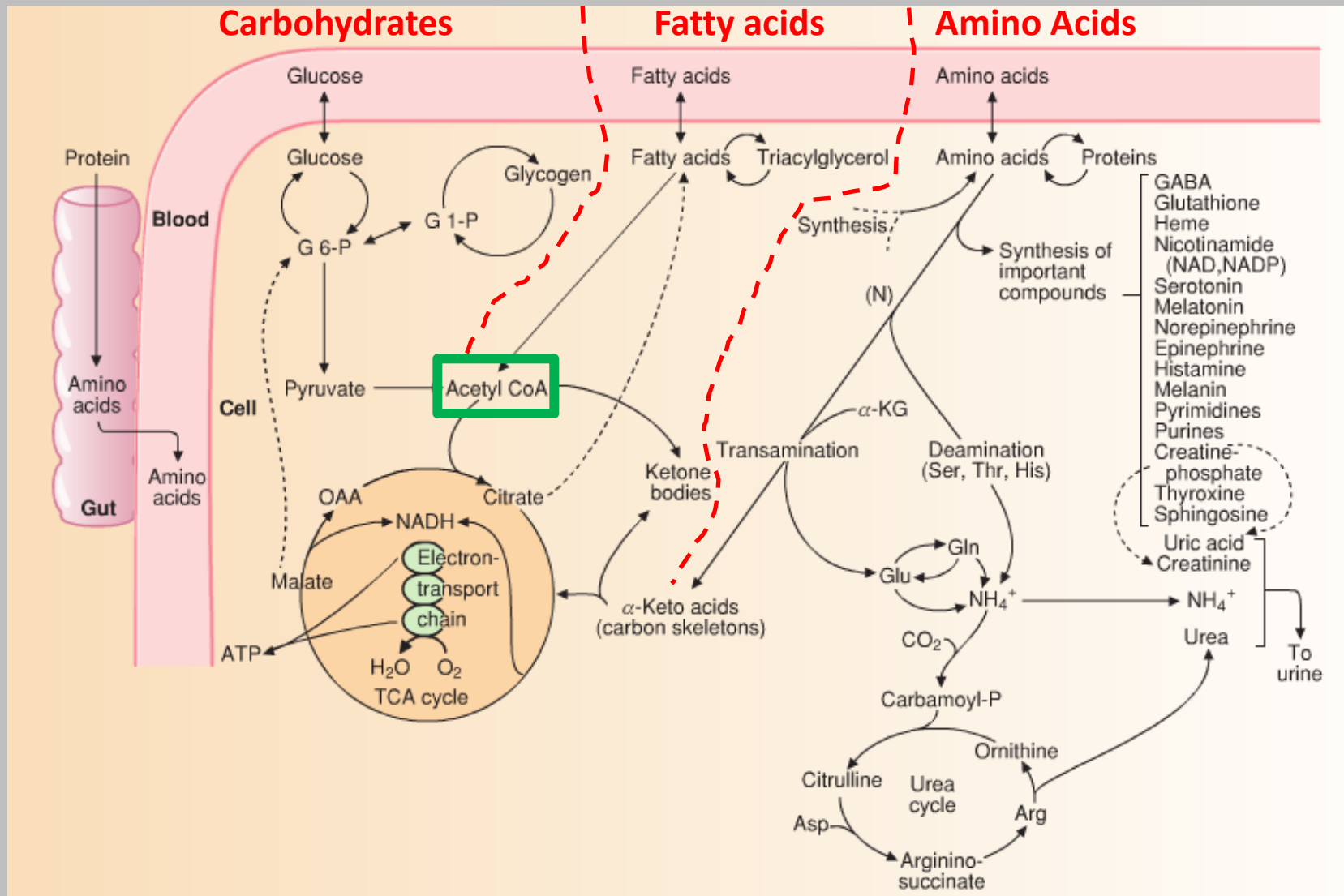


FIG. VII.2. Overview of nitrogen metabolism. The metabolism of nitrogen-containing compounds is shown on the right and that of glucose and fatty acids is shown on the left. This figure shows a hypothetical, composite cell. No single cell type has all of these pathways. Many of the pathways shown are described in the next few chapters. α -KG, α -ketoglutarate; OAA, oxaloacetate; G 6-P, glucose 6-phosphate; G 1-P, glucose 1-phosphate.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>